

DIGITAL SENTINEL

AI-Based Cyberbullying Prevention & Digital Safety Platform

1. Title & Overview

This project focuses on developing **Digital Sentinel**, an intelligent AI-powered web application designed to prevent cyberbullying and online harassment among teenagers and young adults.

The system proactively scans posts and comments in real-time, warns users before harmful content is published, filters abusive messages for recipients, detects repeat offenders, and provides an AI assistant to promote safe online behavior.

The platform is built to support students, schools, moderators, and digital communities by offering a centralized AI-driven safety mechanism that combines prevention, awareness, and policy enforcement.

2. Objective

The primary objective of the **Digital Sentinel** project is to create a safer online environment through AI-based content moderation and awareness tools.

Key objectives include:

- Detecting harmful content (toxicity, harassment, hate speech, threats) in real time.
- Warning users before posting inappropriate messages.
- Automatically filtering abusive content for recipients.
- Identifying repeat offenders using behavioral scoring.
- Providing an AI-powered safety assistant for awareness and emotional support.
- Creating a transparent moderation and digital policy system.
- Reducing cyberbullying incidents among teenagers and young adults.

3. Tools & Technologies Used

The website was built using AI-assisted development tools and modern web technologies.

AI Tools Used:

- **OpenAI**: Content moderation and AI assistant.
- **Wix vibe**: Website creation and growth platform.
- **ChatGPT**: Prompt engineering and information retrieval.
- **GitHub**: Repository management and deployment.

Technologies Used:

- React / Next.js – Frontend development
- Tailwind CSS – UI styling
- FastAPI (Python) – Backend API
- PostgreSQL – Database management
- JWT – Authentication
- Real-time AI moderation API integration

4. Methodology

- **Problem Identification:** Increasing cyberbullying incidents across social media platforms require proactive AI-driven prevention strategies.
- **Planning & Requirement Analysis:** Finalized modules include User Authentication, Social Feed, AI Moderation Engine, Receiver Protection System, Repeat Offender Detection, Admin Dashboard, and AI Awareness Assistant.
- **System Design:** Designed full-stack architecture integrating frontend, backend, AI layer, and database.
- **AI Model Integration:** Implemented real-time moderation scoring across multiple vectors: Toxicity score, Harassment score, Threat score, and Hate speech score.
- **Behavior Scoring System:** Developed violation tracking logic:
 - 3 violations → Warning
 - 5 violations → Moderator alert
 - 10 violations → Temporary suspension
- **User Protection Layer:** Implemented auto-hide harmful messages, block/report options, and a safe-view toggle.
- **Testing & Optimization:** Focused on false-positive reduction, UI responsiveness testing, and performance optimization.
- **Final Deployment:** Hosted on secure cloud infrastructure.

5. Output

The output is a fully functional AI-powered web application featuring:

- Real-time AI content moderation.
- Pre-post warning system.
- Auto-filtering of abusive messages.
- Repeat offender detection logic.
- Admin moderation dashboard.

- Behavioral safety scoring.
- AI chatbot for digital awareness and guidance.
- Transparent violation history & appeal system.

6. Result

Digital Sentinel successfully demonstrates how AI can proactively reduce cyberbullying rather than reacting after damage is done. Users experience reduced exposure to abusive content, increased awareness about safe digital behavior, clear moderation policies, and emotional support guidance via the AI assistant.

The project showcases how AI-driven prevention, behavioral analytics, and transparent moderation can create safer online communities.

7. Conclusion

Digital Sentinel illustrates how intelligent moderation systems can transform digital safety ecosystems. Unlike traditional moderation systems used by platforms such as Meta, TikTok, and YouTube, this system focuses on preventive warnings, behavioral education, and transparency rather than only reactive content removal.

The platform provides a scalable, ethical, and socially impactful solution for schools, digital communities, and online platforms.

8. Project URL

<https://my-site-09inxr0o-asrafathimabegum12.wix-vibe-site.com>

9. GitHub Repository

<https://github.com/asrafathimabegum12-cloud>

<https://github.com/asrafathimabegum12-cloud/cyberbullying-and-online-harassment-on-social-media-platforms->